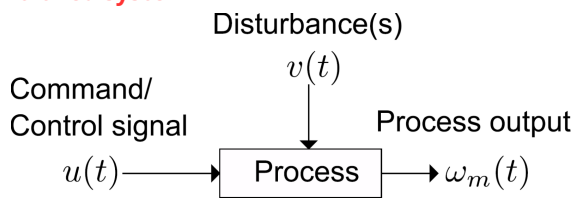
I. Control Theroy and Automation

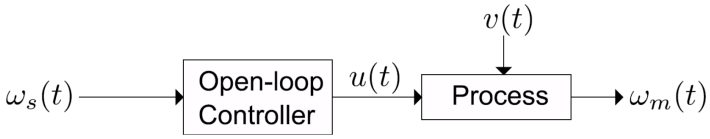
A control system regulates a process using sensors, controllers, actuators, and feedback.

Control theory designs controllers to regulate and optimize dynamic systems.

Uncontrolled system



Open-loop controlled system



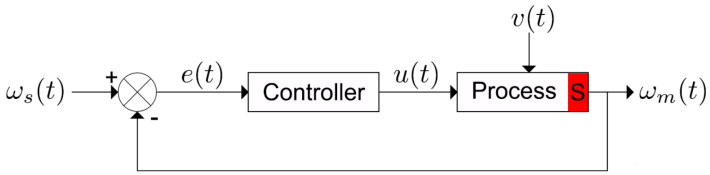
Advantages

- Simple structure: no sensors or controllers needed
- Low cost: fewer components and less complexity

Disadvantages

- No adaptation: cannot correct changes or disturbances
- Unstable performance: quality may decrease over time

Closed-loop controlled system



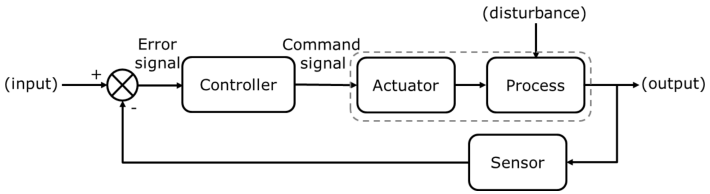
Advantages

- High accuracy: continuous correction improves precision, performance, and efficiency
- Adaptation: reacts to load changes, disturbances, or operating conditions

Disadvantages

- Complex structure: sensors, controllers, and feedback are required
- Higher cost: more components increase total cost
- More maintenance: components need calibration and maintenance

Extended block diagram



Components

Controller

Input: error $e(t)$
Output: command signal $u(t)$

Actuator

Input: command signal $u(t)$
Output: process input signal

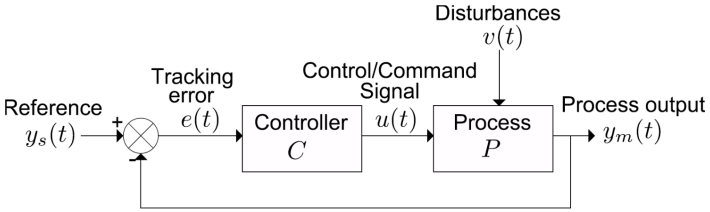
Process

Input: process input signal
Unwanted output: disturbance/noise $v(t)$
Output: actual process output $y_m(t)$

Sensor

Input: actual process output $y_m(t)$
Output: signal to be compared to the desired output signal

Feedback loop in a closed-loop control



$$e(t) = y_s(t) - y_m(t)$$
$$u(t) = C(e(t))$$
$$y_m(t) = P(u(t), v(t))$$

$y_s(t), e(t), u(t), y_m(t)$: Signals
P, C : Systems

Signals vs System

Signals

- Carry information through variations in a physical quantity
- Described as functions of one or more variables
- Can be natural or artificial and are used in communication, measurement, and control

System

- Interconnection of components used to perform a specific function
- Takes input signals, processes or transforms them, and produces output signals
- Describes the relationship between inputs and outputs

Automation pyramid

Level 0. Field/process level
Level 1. Control devices level
Level 2. Supervisory control level
Level 3. Manufacturing planning and execution system (MES)
Level 4. Management/Business level or ERP
Level 5. Digital transformation and AI-driven intelligent ecosystems (cloud level)

Control engineering

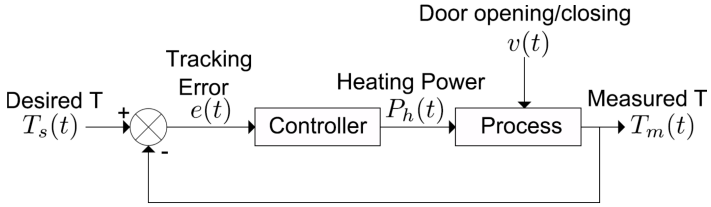
Levels 0-1: real-time control of physical processes using sensors, measurement systems, actuators, and controllers

Automation

Levels 2-5: integration, supervision, optimization, planning, and management of production and business processes

II. Good controllers

Sauna example, closed-loop control



Disturbance rejection

- Ability to reduce the effect of internal or external disturbances on the system output
- Keeps the output close to the desired reference

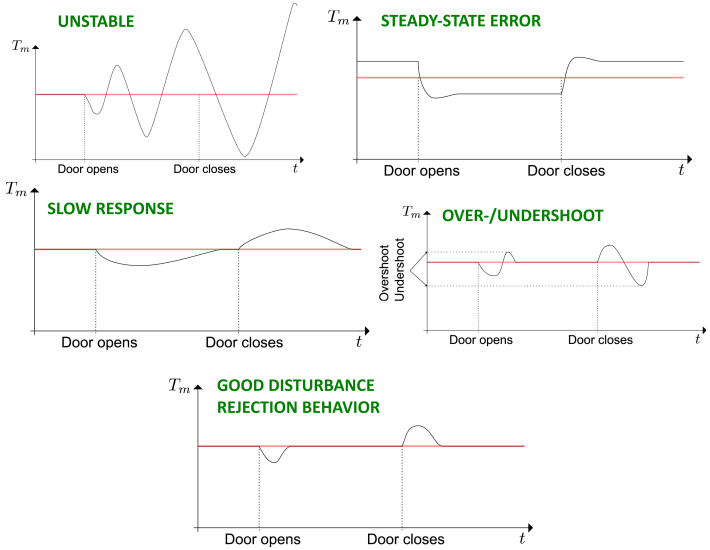
Characteristics

- Disturbance types: internal component changes or external environmental changes
- Effectiveness: controller detects and compensates disturbances
- Control strategies: feedback control reacts after disturbance; feedforward control acts before it affects the output

Performance metrics

- Stability: system remains robust under disturbances
- Steady-state error: final deviation from the reference is minimized
- Overshoot/undershoot: excessive output deviations are reduced
- Rise/settling time: response is fast and stable

Graphical representation



Reference tracking

- Ability to follow the desired reference signal accurately
- Tracking error: difference between reference and output

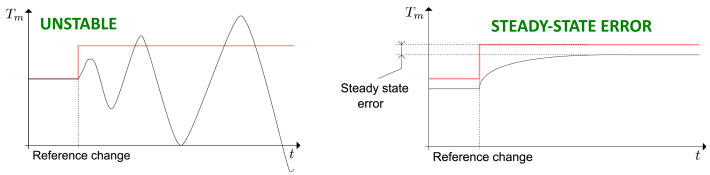
Characteristics

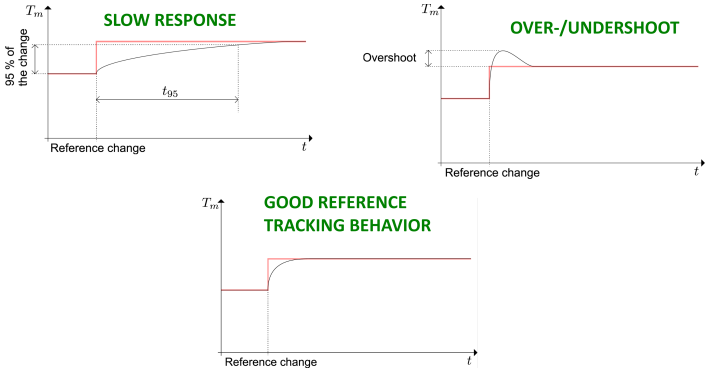
- Tracking accuracy: output follows the reference with minimal deviation
- Effectiveness: controller reduces tracking error under different conditions
- Control strategies: feedback corrects deviations; feedforward anticipates reference changes

Performance metrics

- Stability: tracking remains consistent and robust
- Steady-state error: long-term deviation from the reference is minimized
- Overshoot/undershoot: excessive deviations during tracking are reduced
- Rise/settling time: output reaches the reference quickly and stably

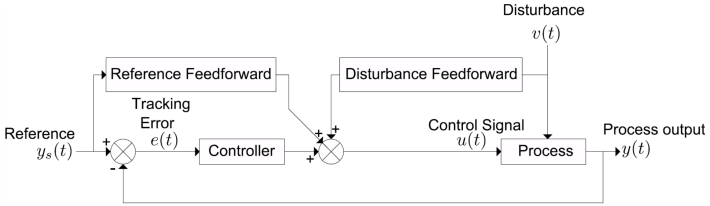
Graphical representation





Feedforward control

- Predictive control strategy that compensates known influences before they affect the output
- Improves performance by anticipating changes instead of only reacting to errors



Reference feedforward

- Compensates reference changes before they affect the system
- Improves tracking by adjusting the control input in advance

Disturbance feedforward

- Compensates measurable disturbances before they affect the output
- Uses a precomputed correction based on the expected disturbance effect

III. Process model development

Damped mass-spring example

Differential equation

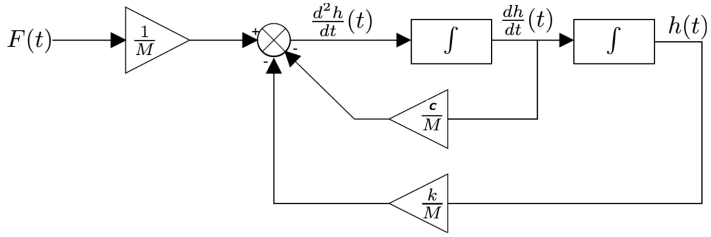
Step 1: Set the differential equation

$$M\frac{d^2h}{dt^2} + c\frac{dh}{dt} + kh = F$$

Step 2: Isolate the term with the highest differential order

$$\frac{d^2h}{dt^2} = \frac{F}{M} - \frac{kh}{M} - \frac{c}{M}\frac{dh}{dt}$$

ODE diagram



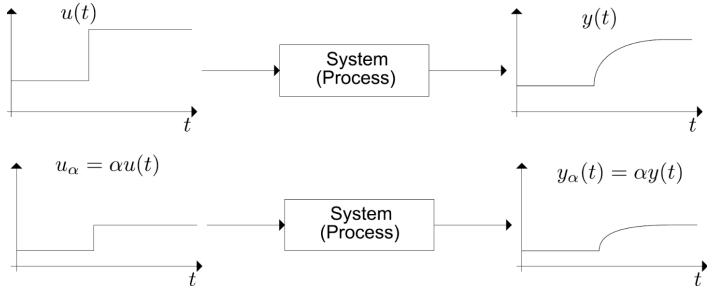
IV. Signals and Systems

Homogeneity (scaling principle)

Scaling the input by a constant also scales the output by the same constant:

- If $u(t) \xrightarrow{h} y(t)$, then $\alpha u(t) \xrightarrow{h} \alpha y(t)$
- If $y(t) = h(u(t))$, then $h(\alpha u(t))$, that is, $h(\alpha u(t)) = \alpha h(u(t))$

Graphical visualization



Homogeneity proof examples

Homogeneous system

$$u(t) \xrightarrow{h} y(t) = 4u(t)$$
$$h(\alpha u(t)) = 4(\alpha u(t)) = \alpha(4u(t)) = \alpha h(u(t))$$
$$\alpha u(t) \xrightarrow{h} \alpha y(t) \qquad \square$$

Non-homogeneous system

$$u(t) \xrightarrow{h} y(t) = u^3(t)$$
$$h(\alpha u(t)) = (\alpha u(t))^3 = \alpha^3 y(t) \neq \alpha y(t)$$
$$h(\alpha u(t)) \neq \alpha h(u(t)) \qquad \square$$

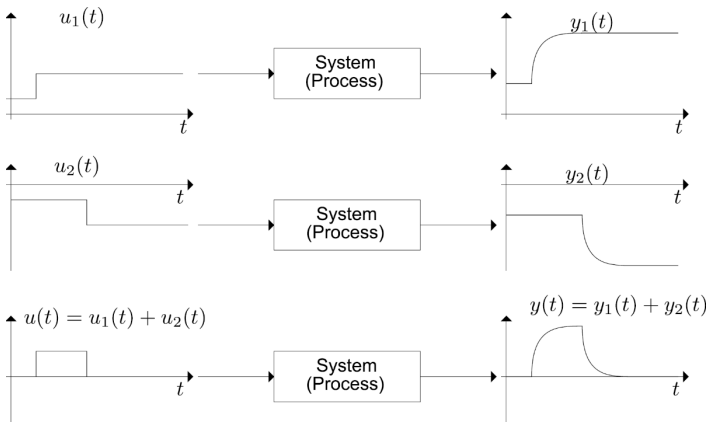
Additivity (superposition principle)

The output of summed inputs equals the sum of the individual outputs:

- If $u_1(t) \to y_1(t)$ and $u_2(t) \to y_2(t)$, then
$$u_1(t) + u_2(t) \to y_1(t) + y_2(t)$$
- If $y_1(t) = h(u_1(t))$ and $y_2(t) = h(u_2(t))$, then
$$h(u_1(t) + u_2(t)) = y_1(t) + y_2(t)$$

That is
$$h(u_1(t) + u_2(t)) = h(u_1(t)) + h(u_2(t))$$

Graphical visualization



Additivity proof examples

Additive system

$$u_1(t) \xrightarrow{h} y_1(t) = 2u_1(t)$$
$$u_2(t) \xrightarrow{h} y_2(t) = 2u_2(t)$$
$$h\big(u_1(t) + u_2(t)\big) = 2\big(u_1(t) + u_2(t)\big)$$
$$= 2u_1(t) + 2u_2(t) = y_1(t) + y_2(t)$$
$$u_1(t) + u_2(t) \xrightarrow{h} y_1(t) + y_2(t) \qquad \square$$

Non-additive system

$$u(t) \xrightarrow{h} y(t) = \sin(u(t))$$
$$h(u_+(t) + u_2(t)) = \sin(u_1(t) + u_2(t)) \neq \sin(u_1(t)) + \sin(u_2(t)) \qquad \square$$

Linear system

A system is linear if it satisfies both the homogeneity (scaling) and additivity principles.

Time-invariant system

System behavior does not change over time. A time-shifted input produces the same time-shifted output:

- If $u(t) \to y(t)$, then
$$u(t - t_0) \to y(t - t_0)$$
- If $y(t) = h(u(t))$, then
$$h(z(t - t_0)) = y(t - t_0)$$

Time-invariance proof example

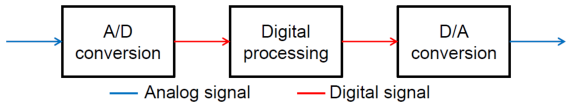
Time-invariant system

$$u(t) \xrightarrow{h} y(t) = u(t) + 7$$
$$h\big(u(t - t_0)\big) = u(t - t_0) + 7 = y(t - t_0)$$
$$u(t - t_0) \xrightarrow{h} y(t - t_0) \qquad \square$$

Time-variant system

$$u(t) \xrightarrow{h} y(t) = u(t) + t$$
$$h\big(u(t - t_0)\big) = u(t - t_0) + t$$
$$y(t - t_0) = u(t - t_0) + (t - t_0)$$
$$h\big(u(t - t_0)\big) \neq y(t - t_0) \qquad \square$$

Digitalization



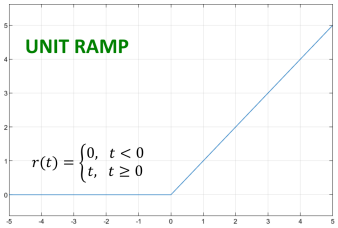
Time discretization

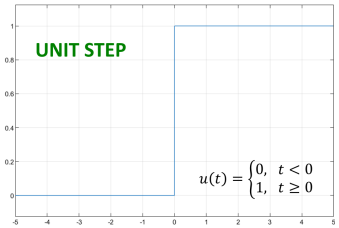
- Sampling: continuous signal measured at fixed time intervals
- Sampling rate: number of samples per second [Hz or kHz]

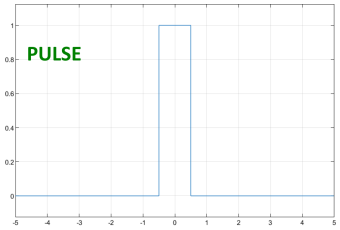
Amplitude discretization

- Quantization: signal values are rounded to discrete levels
- Quantization type: uniform uses equal steps; non-uniform uses variable steps
- Quantization step: coarse steps give lower precision; fine steps give higher precision

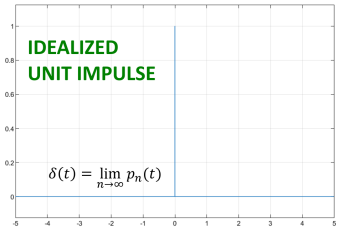
Basic input signals







$$p_n(t) = \begin{cases} \frac{n}{2}, & -\frac{1}{n} < t < +\frac{1}{n} \\ 0, & elsewhere \end{cases}$$



$$\delta(t) = \begin{cases} 1, & t = 0 \\ 0, & elsewhere \end{cases}$$

V. Linearization

Steps for linearizing a function

Step 1. Determine the nonlinear differential equation:
$$M\ddot{h} + c\dot{h} = Mg - kh^3$$

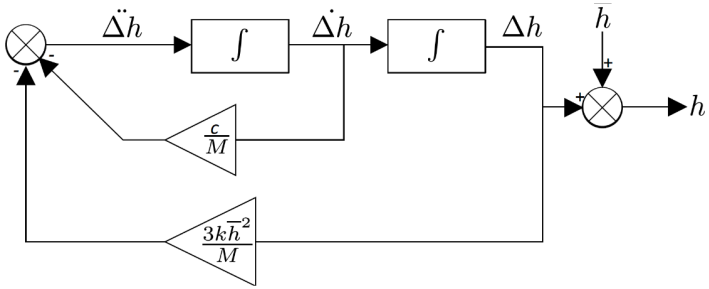
Step 2. Define $f(\ddot{h}, \dot{h}, h) = 0$:
$$M\ddot{h} + c\dot{h} + kh^3 - Mg = 0$$

Step 3. Define the slopes Δ:
$$\Delta\ddot{h} = \ddot{h} - \bar{\ddot{h}}$$

Step 4. Calculate the partial derivatives of f at the steady state for all the variables:
$$\frac{\partial f}{\partial \ddot{h}} = M\Delta\ddot{h}$$

Step 5. Establish a linear equation using the slopes solving for the highest differential order:
$$\Delta\ddot{h} = -\frac{c}{M}\Delta\dot{h} - \frac{3k(\bar{h})^2}{M}\Delta h$$

ODE diagram



VI. Controllers

Analog controller

- Processes continuous signals using analog components, e.g. resistors, capacitors, and operational amplifiers
- Control signal: continuous voltage or current
- Main feature: fast real-time response with simple control logic

Digital controller

- Uses A/D and D/A converters → discretization and delays
- Slightly reduced dynamic performance due to discretization
- Flexible, precise, scalable, and easy to maintain
- Compact and cost-effective hardware
- Supports multitasking, noise filtering, and advanced algorithms

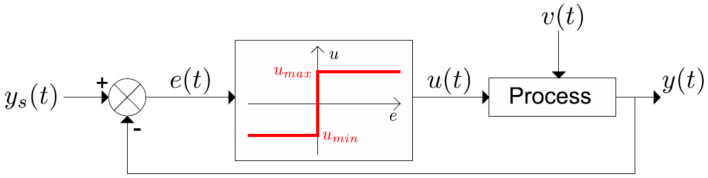
Implementation platforms

- IPCs: complex real-time industrial control
- MCUs: simple low-cost embedded control
- PLCs: robust automation in harsh environments
- DSPs: fast real-time signal calculations
- FPGAs: ultra-fast parallel control
- SBCs: prototyping and low-cost control
- Cloud: remote monitoring, control, and data analysis

Architecture and communication

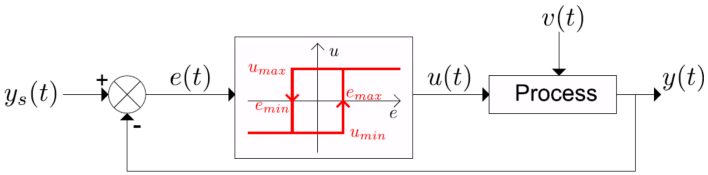
- Centralized control: one unit handles processing, conversion, inputs, and outputs; simple but less scalable
- Fieldbus communication: distributed devices communicate via industrial networks; scalable and modular
- Examples: ProfiNET, CAN/CANopen, EtherCAT

2-point controller



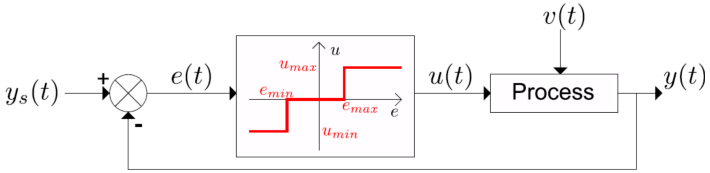
$$\begin{cases} u = u_{max} & \text{if } e > 0 \\ u = u_{min} & \text{if } e < 0 \end{cases}$$

2-point controller with hysteresis



$$\begin{cases} u = u_{max} & \text{if } e > e_{max} \\ u = u_{max} & \text{if } e_{min} < e < e_{max} \text{ and } \dot{e} < 0 \\ u = u_{min} & \text{if } e_{min} < e < e_{max} \text{ and } \dot{e} > 0 \\ u = u_{min} & \text{if } e < e_{min} \end{cases}$$

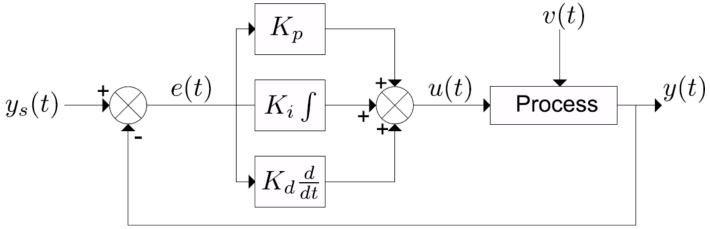
3-point controller



$$\begin{cases} u = u_{max} & \text{if } e > e_{max} \\ u = 0 & \text{if } e_{min} < e < e_{max} \\ u = u_{min} & \text{if } e < e_{min} \end{cases}$$

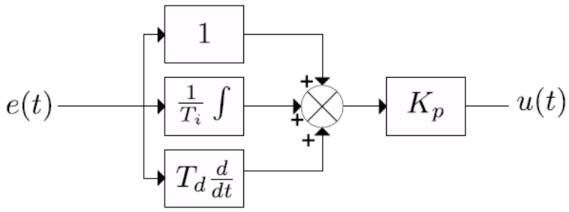
Proportional-Integral-Derivative (PID) controller

Mathematical parametrization



$$u(t) = K_p(e(t)) + K_i \int_0^t e(\tau)\, d\tau + K_d \frac{de(t)}{dt}$$

Technical usual parametrization



$$u(t) = K_p \left(e(t) + \frac{1}{T} \int_0^t e(\tau)\, d\tau + T_d \frac{de(t)}{dt} \right)$$

Controller gain	Integral time	Derivative time
K_p	$T_i = \frac{K_p}{K_i}$	$T_d = \frac{K_d}{K_p}$

PID principle

Proportional term (P)

- Immediate correction proportional to current error
- Reduces rise time
- Cannot remove steady-state error alone
- Too high → oscillation/overshoot

Integral term (I)

- Eliminates steady-state error
- Slow reaction
- Large integral action → overshoot (integral wind-up)

Derivative term (D)

- Predictive action: reacts to error rate of change
- Reduces overshoot and oscillations
- Sensitive to noise

PI controller

- Faster than pure I
- Deletes steady-state error
- Mild overshoot

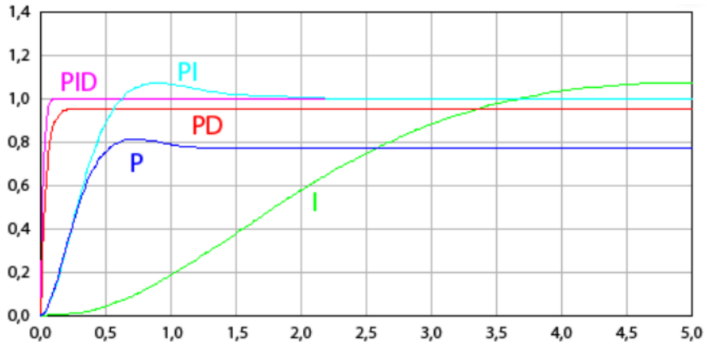
PD controller

- Fast response
- Low overshoot
- Still has steady-state error

PDI controller

- Fast rise time
- No steady-state error
- Limited overshoot
- Best compromise between accuracy + stability

Graphical representation

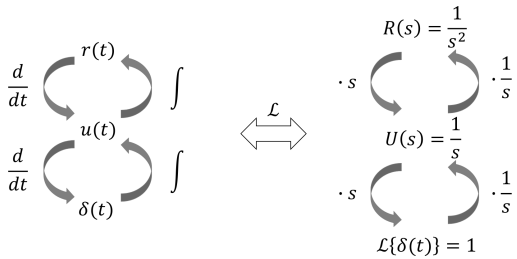


VII. Laplace transform

Laplace transform

The Laplace transform is defined for $t \geq 0$ as

$$\boxed{F(s) = \mathcal{L}\{f(t)\} = \int_0^\infty f(t) \cdot e^{-st}\, dt}$$



Laplace transform of some functions

	$f(t),\ t \geq 0$	$F(s) = \mathcal{L}\{f(t)\}$
Constant	$1 \quad ; \quad a$	$\frac{1}{s} \quad ; \quad \frac{a}{s}$
Step function	$t \quad ; \quad at$	$\frac{1}{s^2} \quad ; \quad \frac{a}{s^2}$
Derivative	$\frac{df(t)}{dt}$	$sF(s) - f(0)$
Integral	$\int_0^t f(\tau)\, d\tau$	$\frac{1}{s}F(s)$
Initial value	$\lim_{t\to 0} f(t)$	$\lim_{s\to\infty} sF(s)$
Steady-state value	$\lim_{t\to\infty} f(t)$	$\lim_{s\to 0} sF(s)$
Delayed input	$f(t - t_0)$	$\exp(-s \cdot t_0)F(s)$
Exponential	$\exp(-at)$	$\frac{1}{s+a}$

Benefits of analysis in the s-domain

- Converts differential equations into algebraic equations
- Defines direct input-output relation using transfer functions
- Enables frequency response analysis
- Simplifies stability and system behavior analysis
- Supports structured open-loop and closed-loop analysis
- Makes transfer functions easier to manipulate

VIII. Transfer functions

$$sY(s) + C(s)Y(s) = G(s)u(s) \Longrightarrow Y(s)\left[s + C(s)\right] = G(s)u(s)$$

$$H_{Y,U}(s) = \frac{Y(s)}{U(s)} = \frac{C(s)}{s + G(s)} = \frac{K_Y}{1 + \tau_Y \cdot s}$$

where K_Y is the gain and τ_Y is the time constant.

Transfer function of the PID controller

ODE of the PID controller:

$$u(t) = K_p\left(e(t) + \frac{1}{T_i}\int_0^t e(\tau)\, d\tau + T_d\dot{e}(t)\right)$$

Transfer function:

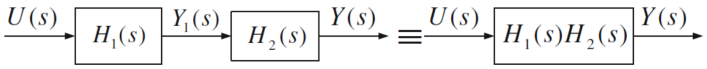
$$U(s) = K_p\left(E(s) + \frac{1}{T_i}\frac{E(s)}{s} + T_d s E(s)\right)$$

$$U(s) = E(s)\left(K_p\frac{T_iT_ds^2 + T_is + 1}{T_is}\right)$$

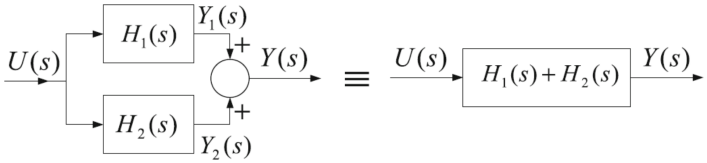
$$C(s) = \frac{U(s)}{E(s)} = \left(K_p\frac{T_iT_ds^2 + T_is + 1}{T_is}\right)$$

Block diagram manipulation

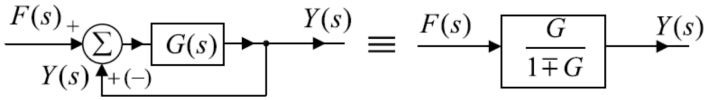
Serial connection



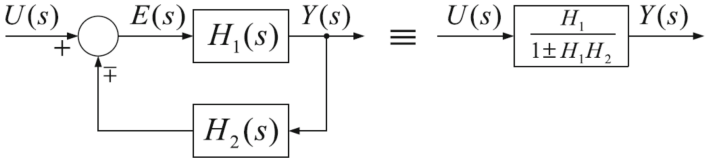
Parallel connection



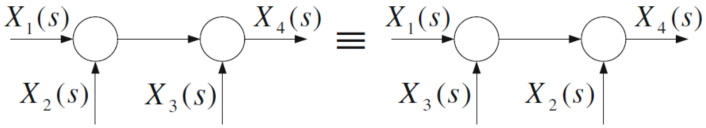
Feedback loop 1



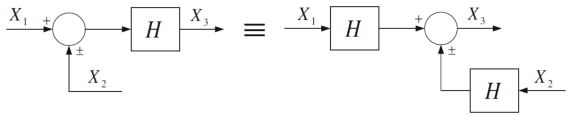
Feedback loop 2



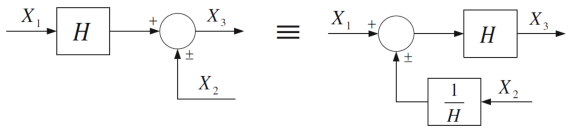
Summation points



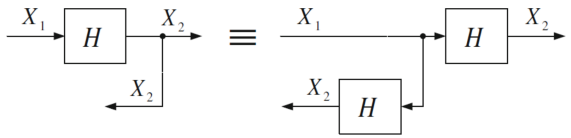
Relocation of a summation 1



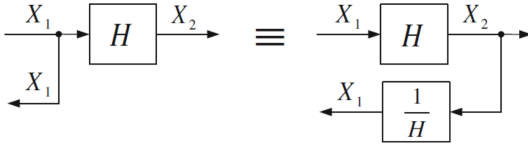
Relocation of a summation 2



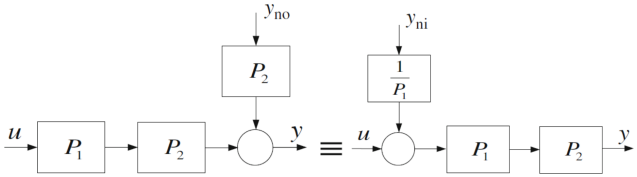
Relocation of a junction 1



Relocation of a junction 2

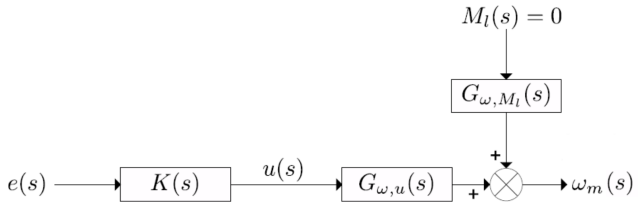


Relocation of a disturbance



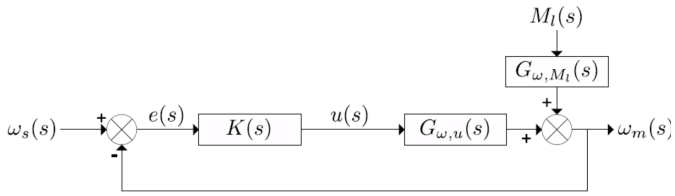
Systems transfer function

Reference open-loop transfer function



$$L(s) = \frac{\omega_m(s)}{e(s)} = K(s)G_{\omega,u}(s)$$

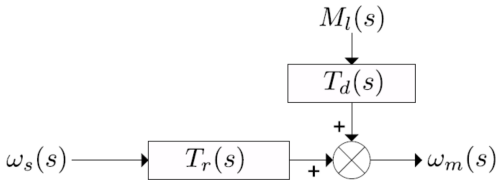
Closed-loop transfer function



$$T_r(s) = \frac{\omega_m(s)}{\omega_s(s)} = \frac{K(s)G_{\omega,u}(s)}{1 + K(s)G_{\omega,u}(s)}$$

$$T_d(s) = \frac{\omega_m(s)}{M_l(s)} = \frac{G_{\omega,M_l}(s)}{1 + K(s)G_{\omega,u}(s)}$$

Controlled system response



$$\omega_m(s) = T_r(s)\omega_s(s) + T_d(s)M_l(s)$$

Zeros and Poles

A system is described by its transfer function:

$$G(s) = \frac{N(s)}{D(s)}$$

- Zeros: roots of $N(s)$
- Poles: roots of $D(s)$

Stability

Bounded input gives bounded output:

$$|u(t)| < \infty \implies |y(t)| < \infty$$

Pole conditions:

- $Re < 0$: stable
- Any $Re > 0$: unstable
- Single pole with $Re = 0$: marginally stable
- Repeated poles with $Re = 0$: unstable

Causality

Output depends only on present and past inputs:

- $\deg(N) < \deg(D)$: strictly causal
- $\deg(N) = \deg(D)$: causal
- $\deg(N) > \deg(D)$: non-causal

Oscillations

Oscillations occur with complex poles:

- Real poles only: no oscillation
- $j \pm (Re = 0)$: sustained oscillation
- $j \pm (Re < 0)$: damped oscillation
- $j \pm (Re > 0)$: increasing oscillation

Final value theorem (steady-state)

$$t \to \infty \overset{\mathcal{L}}{\iff} s \to 0 \quad ; \quad \boxed{\lim_{t\to\infty} y(t) = \lim_{s\to 0} s \cdot Y(s)}$$

Unit step final value

$$Y(s) = G(s)U(S) = G(s) \cdot \frac{1}{s}$$

$$\lim_{s\to 0} s \cdot Y(s) = \lim_{s\to 0} s \cdot \frac{1}{s} \cdot G(s) = \lim_{s\to 0} G(s) = G(s=0)$$

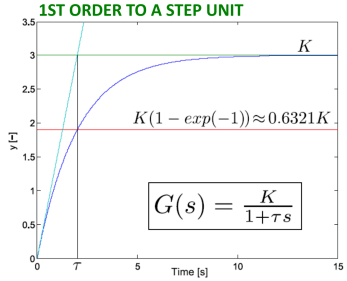
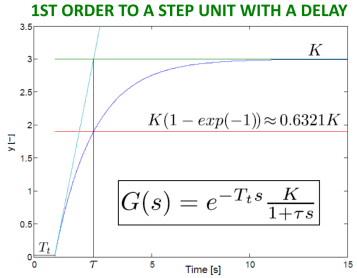
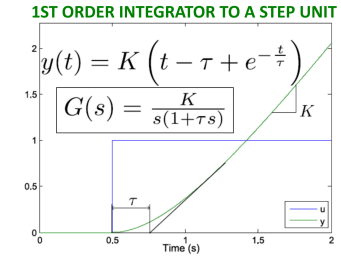
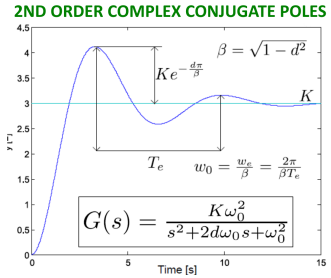
Initial value theorem

$$t \to 0 \overset{\mathcal{L}}{\iff} s \to \infty \quad ; \quad \boxed{\lim_{t\to 0} y(t) = \lim_{s\to\infty} s \cdot Y(s)}$$

Unit step final value

$$Y(s) = G(s)U(S) = G(s) \cdot \frac{1}{s}$$

$$\lim_{s\to\infty} s \cdot Y(s) = \lim_{s\to\infty} s \cdot \frac{1}{s} \cdot G(s) = \lim_{s\to\infty} G(s)$$

IX. First- and second-order responses

Type of lag responses